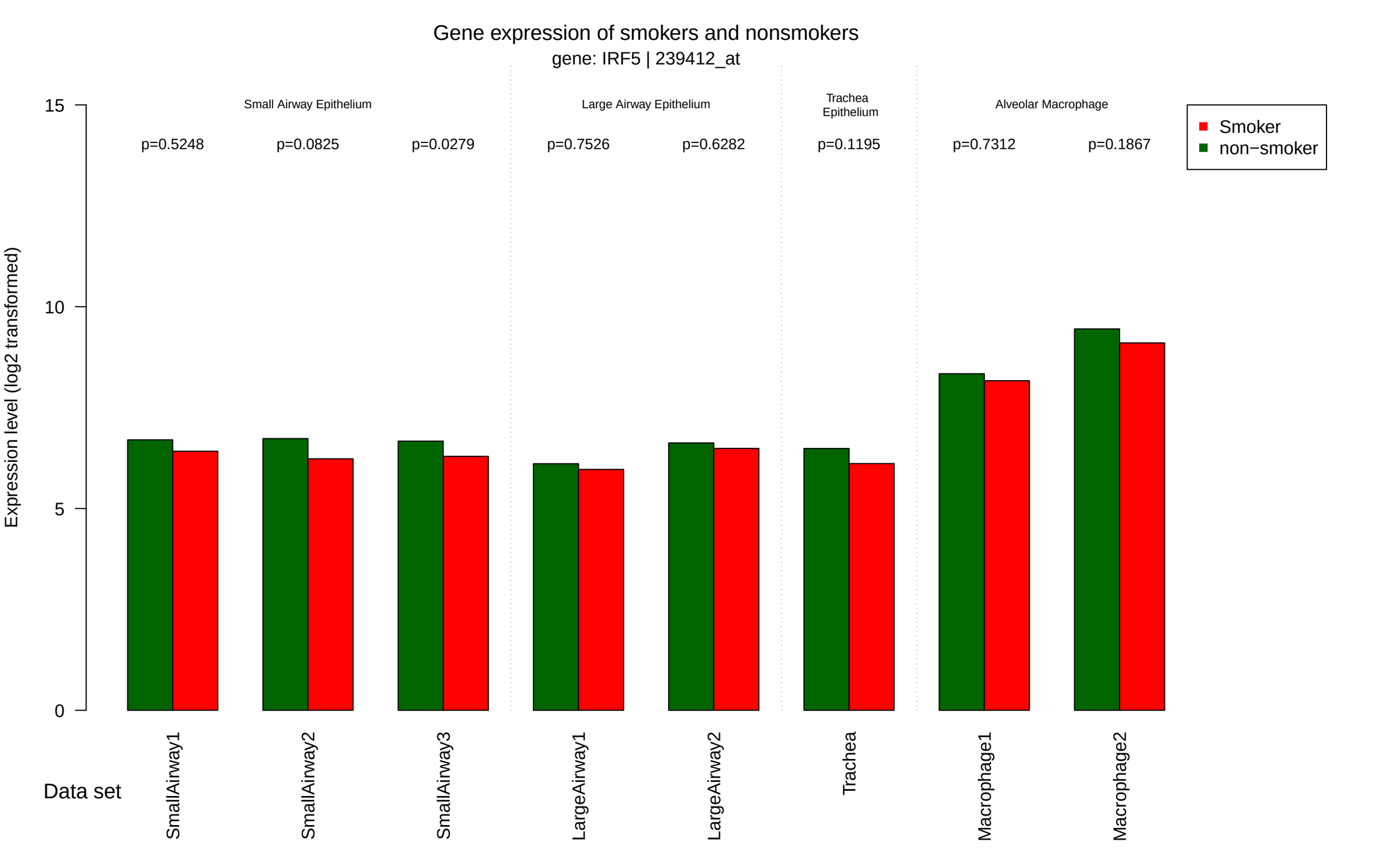
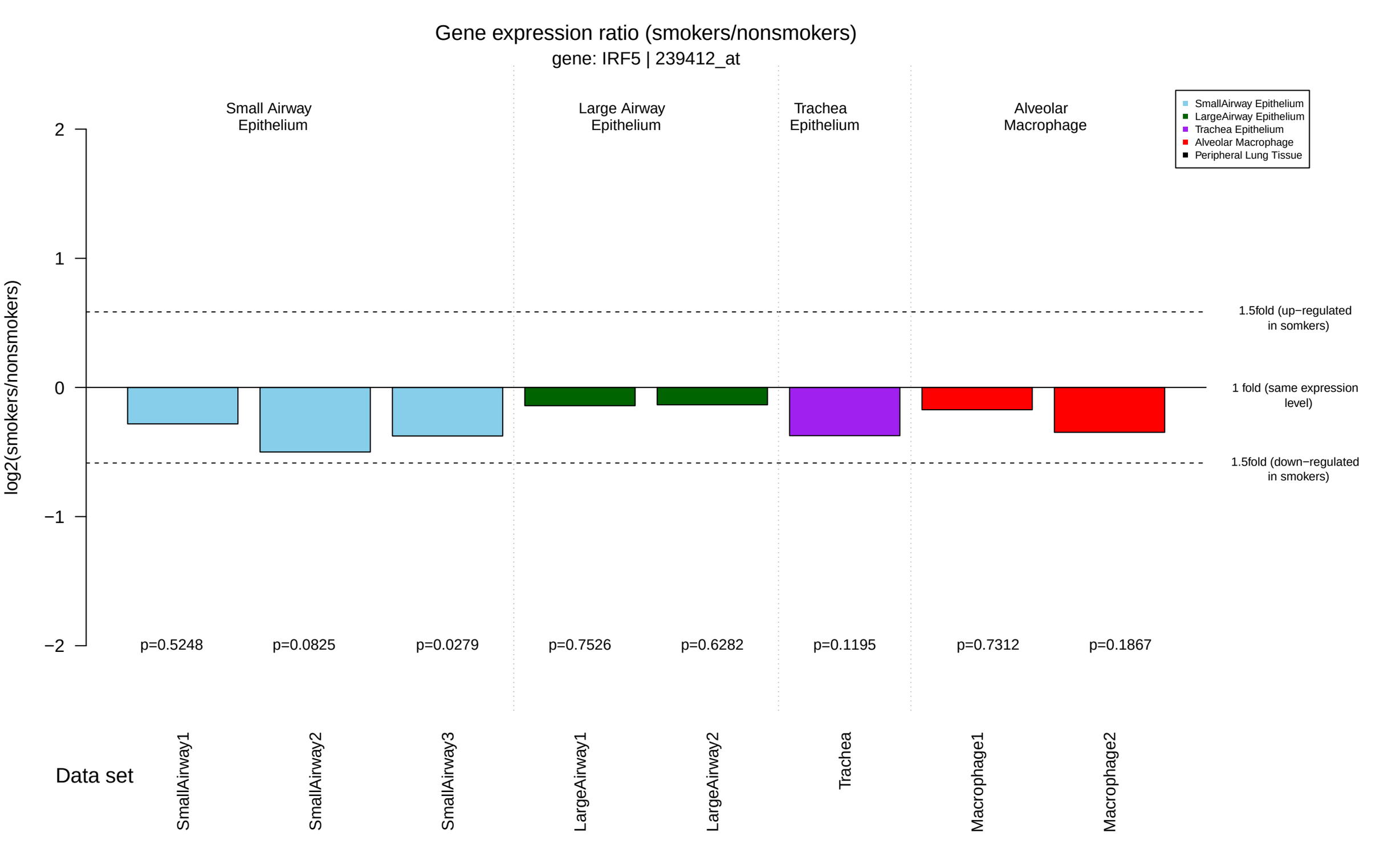
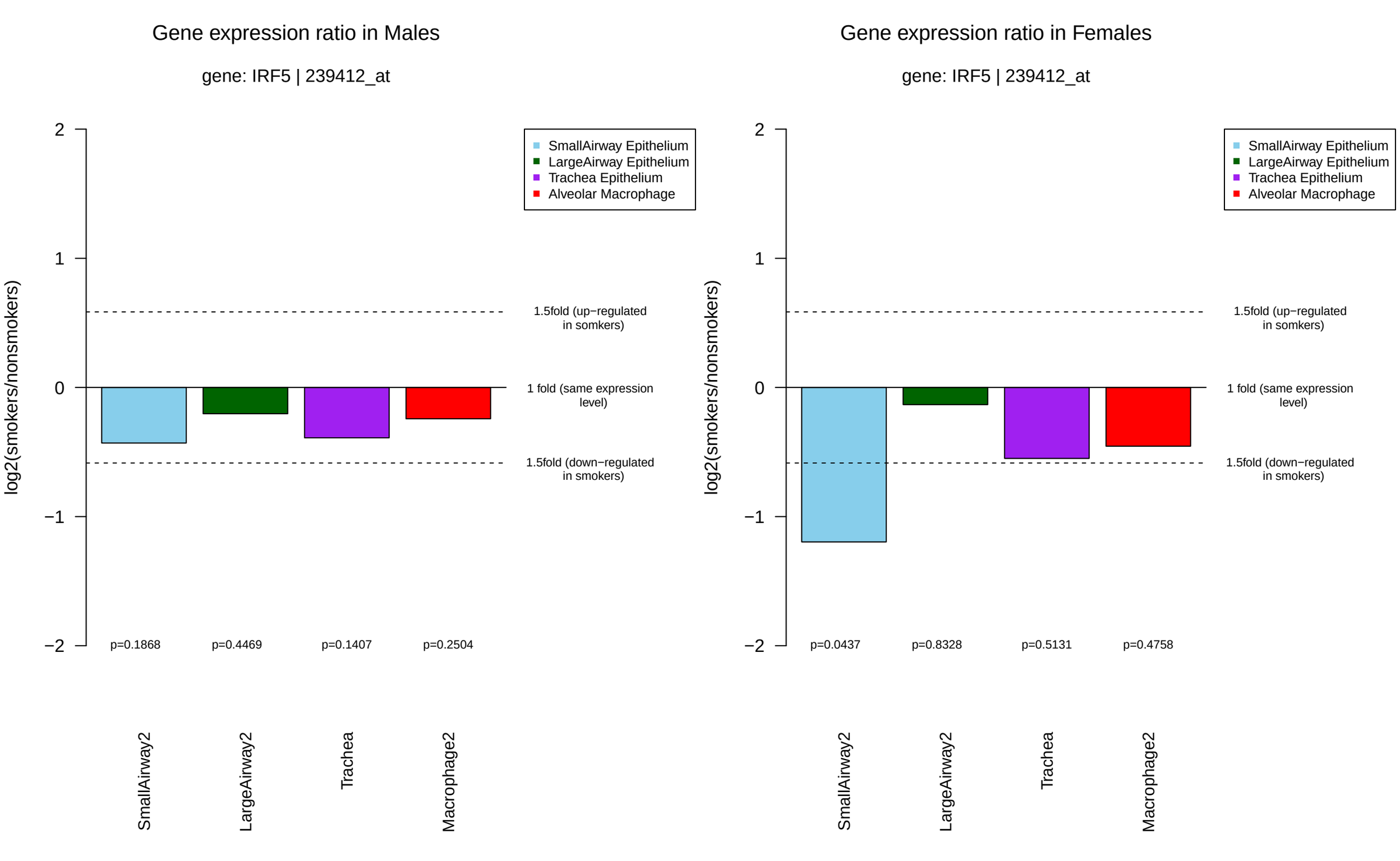
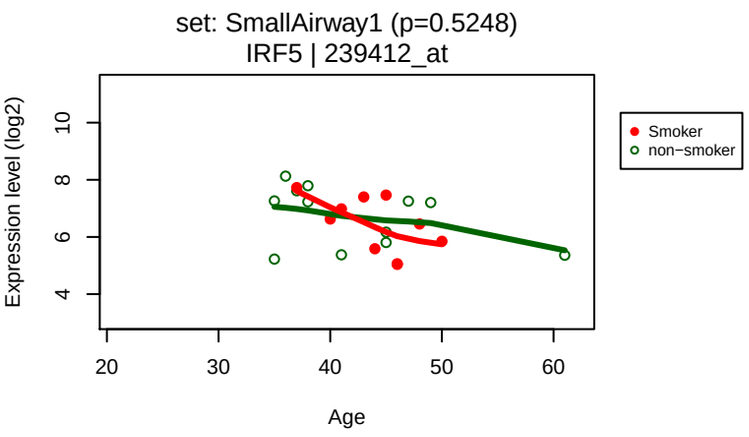
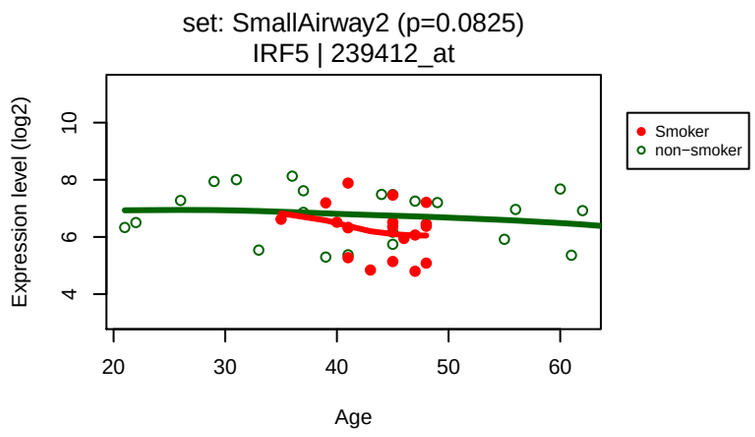
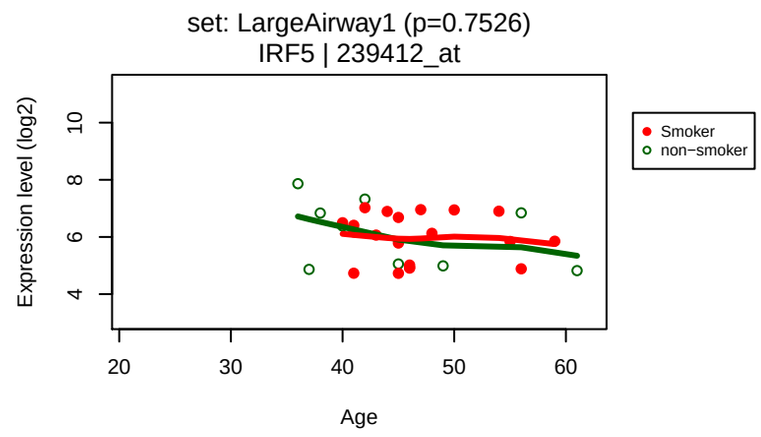

Gene expression of smokers and nonsmokers
gene: IRF5 | 239412_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5248
p=0.0825
p=0.0279
p=0.7526
p=0.6282
p=0.1195
p=0.7312
p=0.1867
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

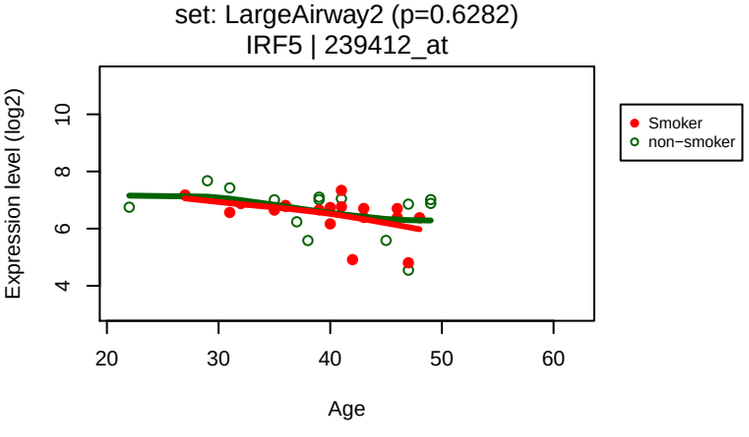

Gene expression ratio (smokers/nonsmokers)
gene: IRF5 | 239412_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5248
p=0.0825
p=0.0279
p=0.7526
p=0.6282
p=0.1195
p=0.7312
p=0.1867
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

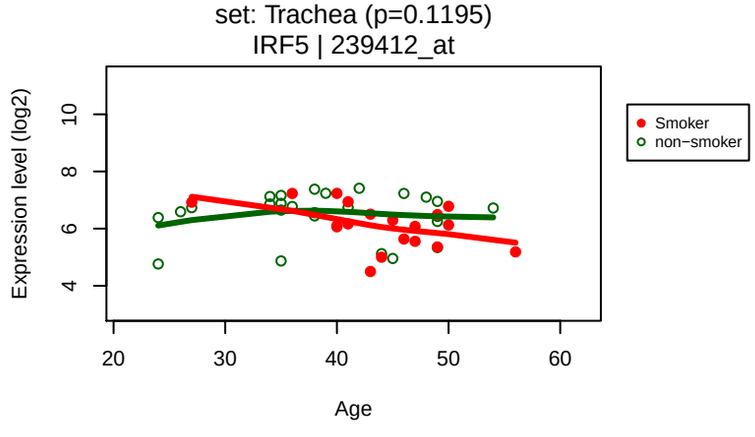

Gene expression ratio in Males
gene: IRF5 | 239412_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1868
p=0.4469
p=0.1407
p=0.2504
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IRF5 | 239412_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0437
p=0.8328
p=0.5131
p=0.4758
SmallAirway2
LargeAirway2
Trachea
Macrophage2

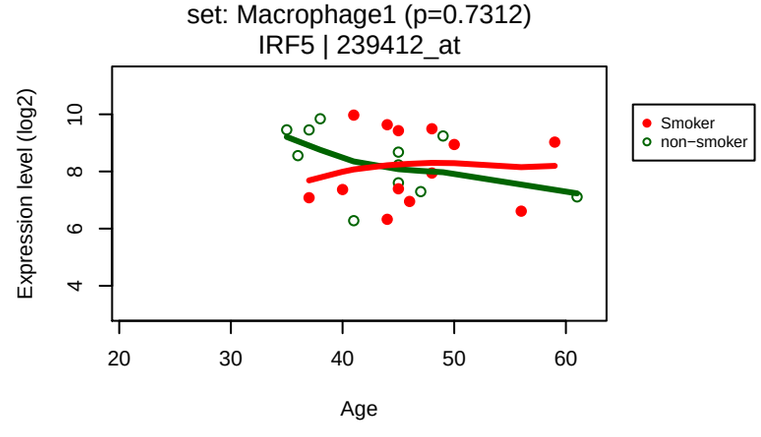

set: SmallAirway1 (p=0.5248)
IRF5 | 239412_at
Expression level (log2)
Age
Smoker
non−smoker

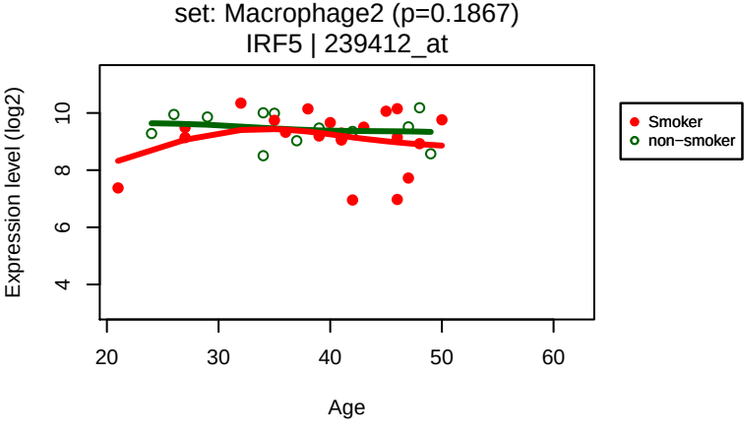

set: SmallAirway2 (p=0.0825)
IRF5 | 239412_at
Expression level (log2)
Age
Smoker
non−smoker

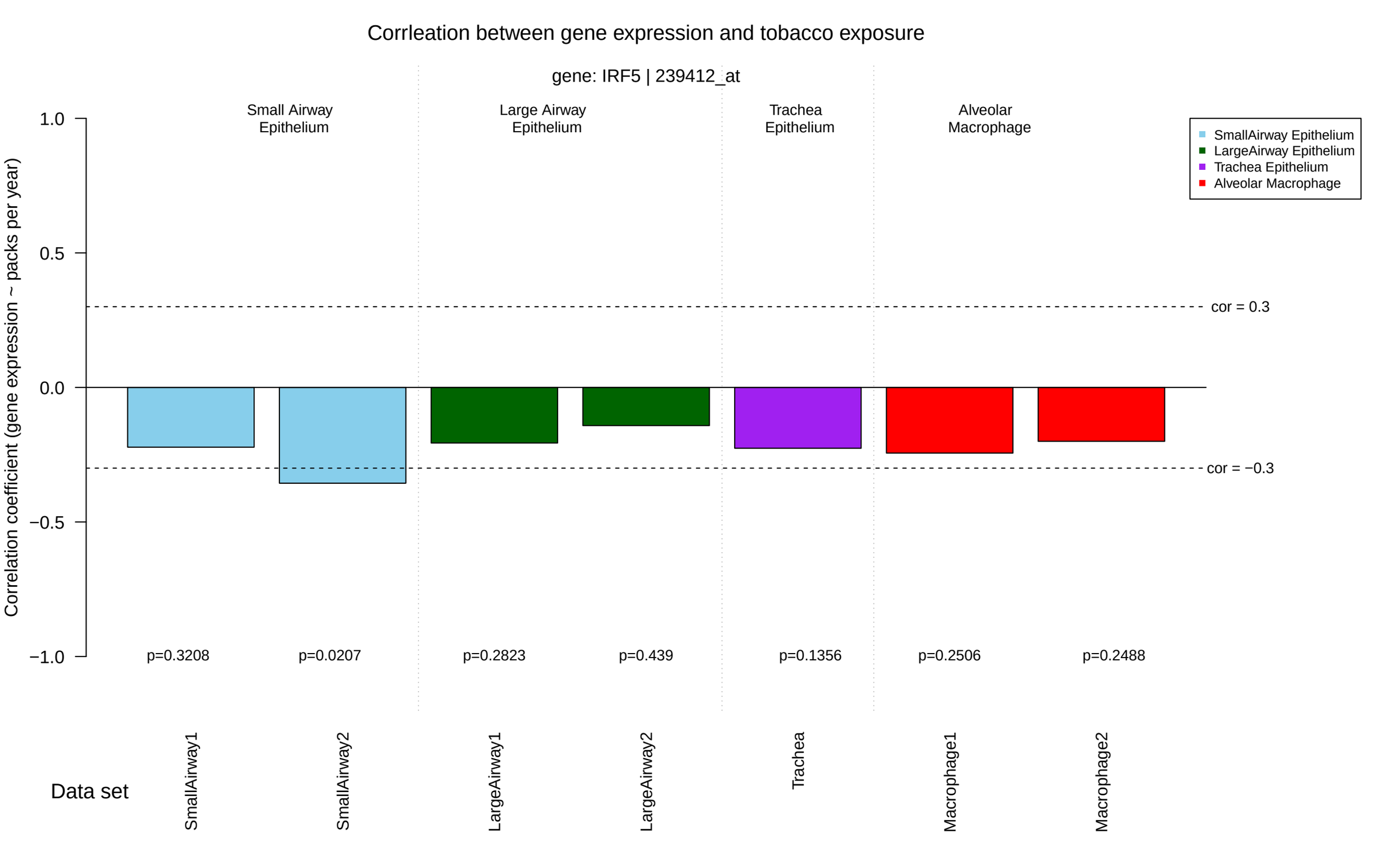

set: LargeAirway1 (p=0.7526)
IRF5 | 239412_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6282)
IRF5 | 239412_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1195)
IRF5 | 239412_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7312)
IRF5 | 239412_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1867)
IRF5 | 239412_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IRF5 | 239412_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3208
p=0.0207
p=0.2823
p=0.439
p=0.1356
p=0.2506
p=0.2488
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2